

Criterion B: Design

Output visualization

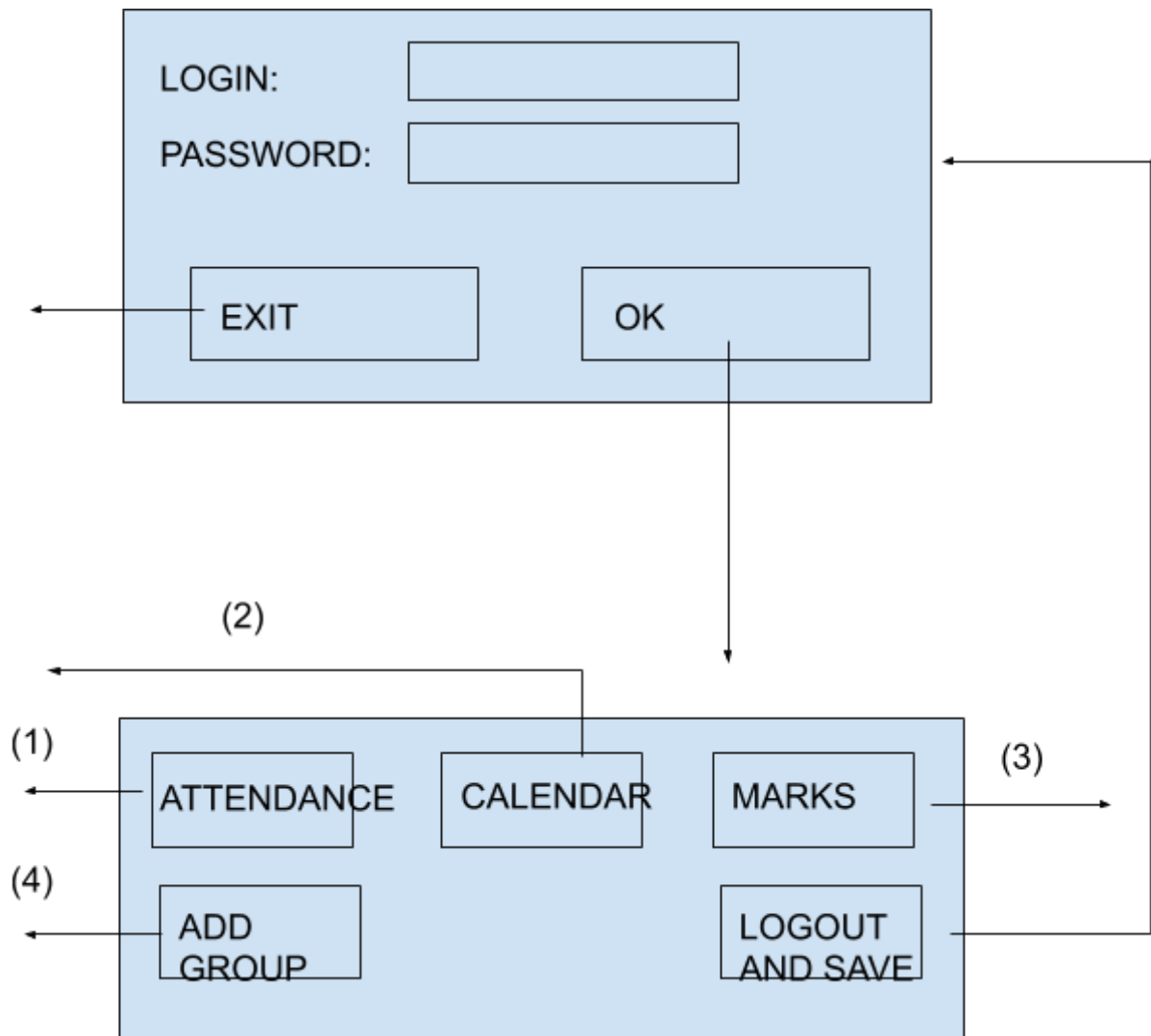


Fig. 1. Login panel and main menu.

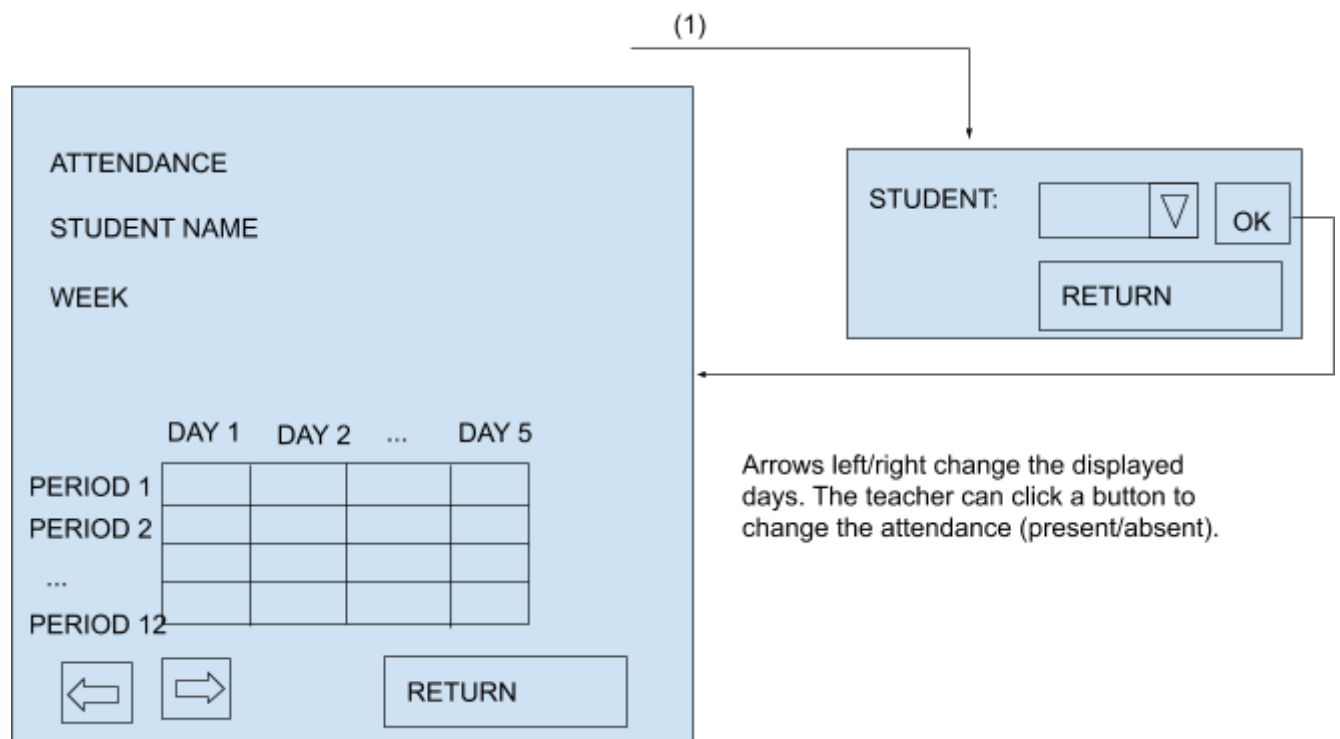


Fig. 2. Attendance panel.

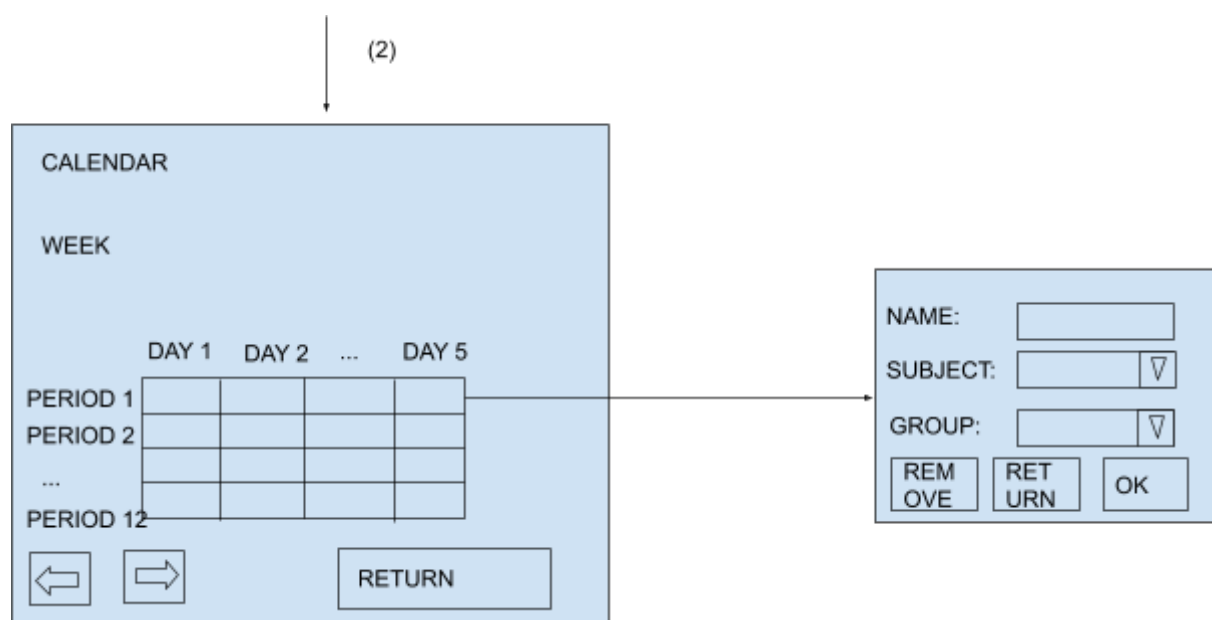


Fig. 3. Calendar panel.

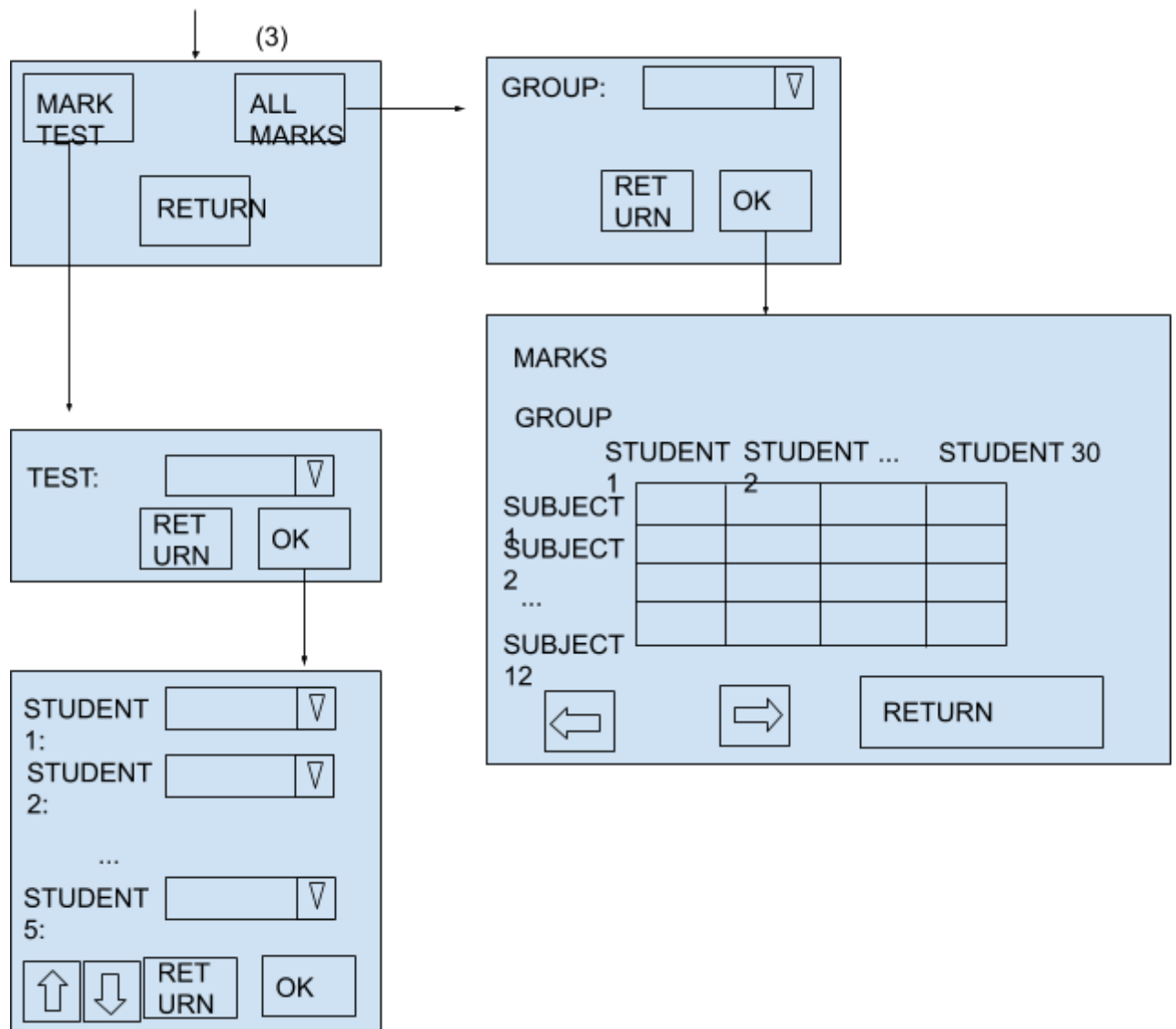


Fig. 4. Marks panel.

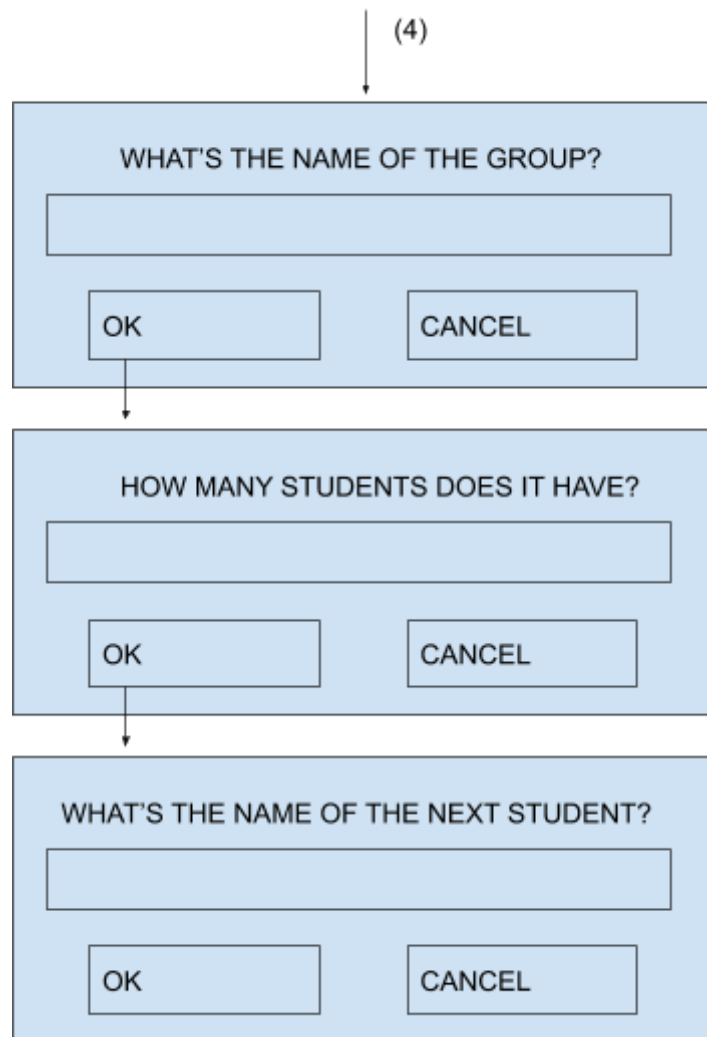


Fig. 5. "Add group" input panels

Input and output data

Panel	Input data	Output data
Login panel	Login, password, button clicks	Main menu panel
Main menu panel	Button clicks	Attendance, calendar, marks and “add group” panels
Attendance panel student choice	Student, button clicks	Student names, attendance panel or main menu
Attendance panel	Attendance, button clicks	Attendance at specific days and periods in one week, student name, week number
Calendar (test) panel	New test, button clicks	Test edition panel, test names, week number
Test edition panel	Test name, subject, group name, button clicks	Subjects, group names
Marks 3-button selection panel	Button clicks	Button labels, next panels
“Mark test” test choice panel	Button clicks, test name	Test names, “mark test” panel
“Mark test” panel	Marks, button clicks	Student names, marks
“All marks” group choice panel	Group name, button clicks	Group names, “all marks” panel
“All marks” panel	Button clicks	Student names, group name, subjects, marks
“Add group” 1st panel	Group name, button clicks	Next panel, main menu
“Add group” 2nd panel	Group size, button clicks	Next panel, main menu
“Add group” 3rd panel	Student name, button clicks	Next panel, main menu

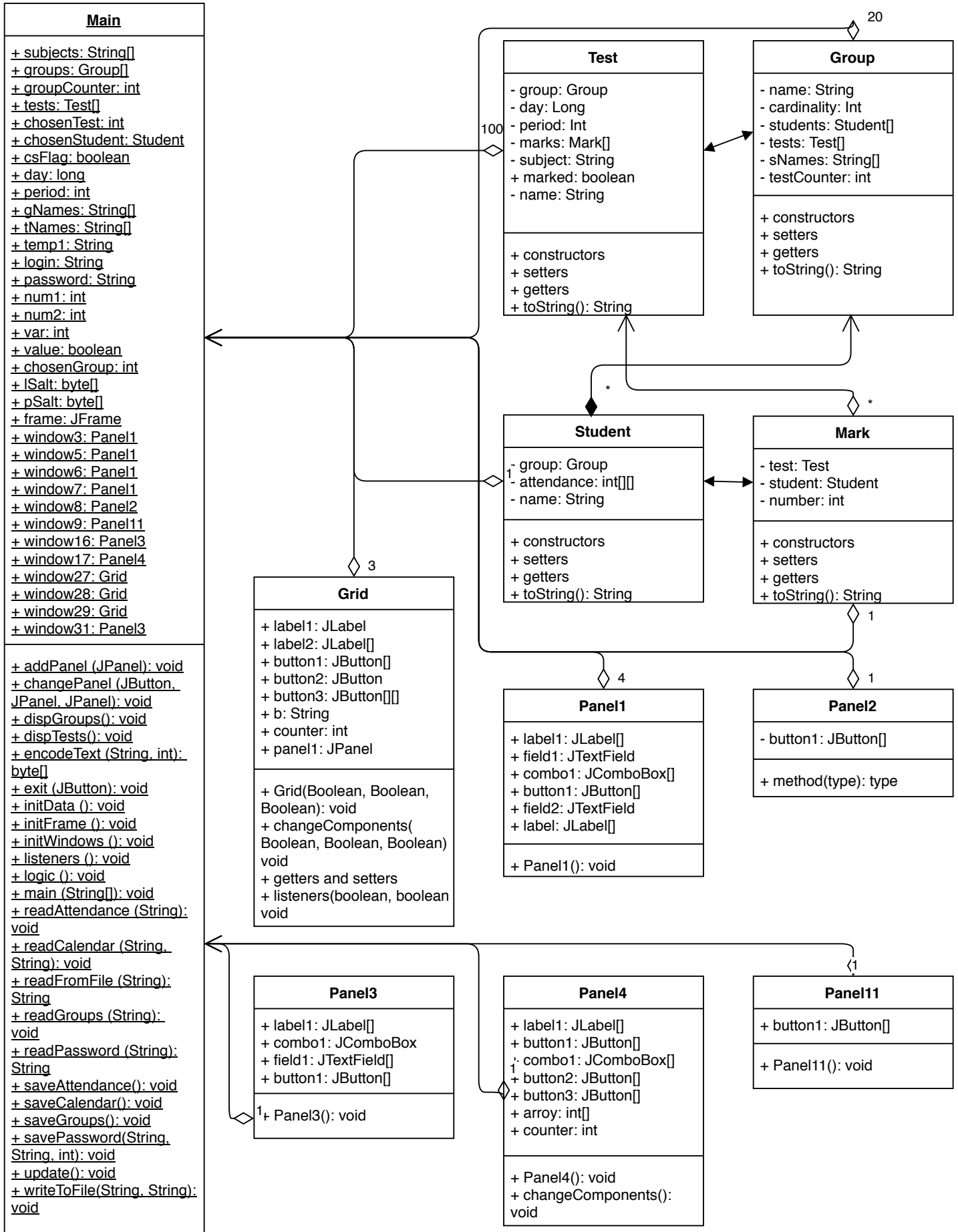
Test plan

Test number	Test Type	Nature of Test	Example
1	Upon starting the program, the login panel comes up	Check that the login panel is correctly displayed	Login panel with "login" and "password" fields
2	In order to proceed to menu, the correct login and password have to be put in	First put in wrong login and password, and then the right ones and see what happens	First use: "apple" as login and "banana" as password; later: "admin" and "password" (the correct combination)
3	The interface is intuitive and there are few buttons	Browse through the panels and see if their number of components is small	Check the login panel, main menu and attendance panel
4	It's possible to add groups and students	It should be possible to use the "new group" button to add a new group and set its name, number of students and students	Add the group "1A" with 6 students: "Anne", "Lucy", "Veronica", "Robert", "James" and "John"
5	It is possible to edit attendance of different students	The attendance panel should make it possible to change attendance upon clicking on any button on the attendance table	Set that "Anne" was absent on Monday and Tuesday. Then make her present again on Monday.

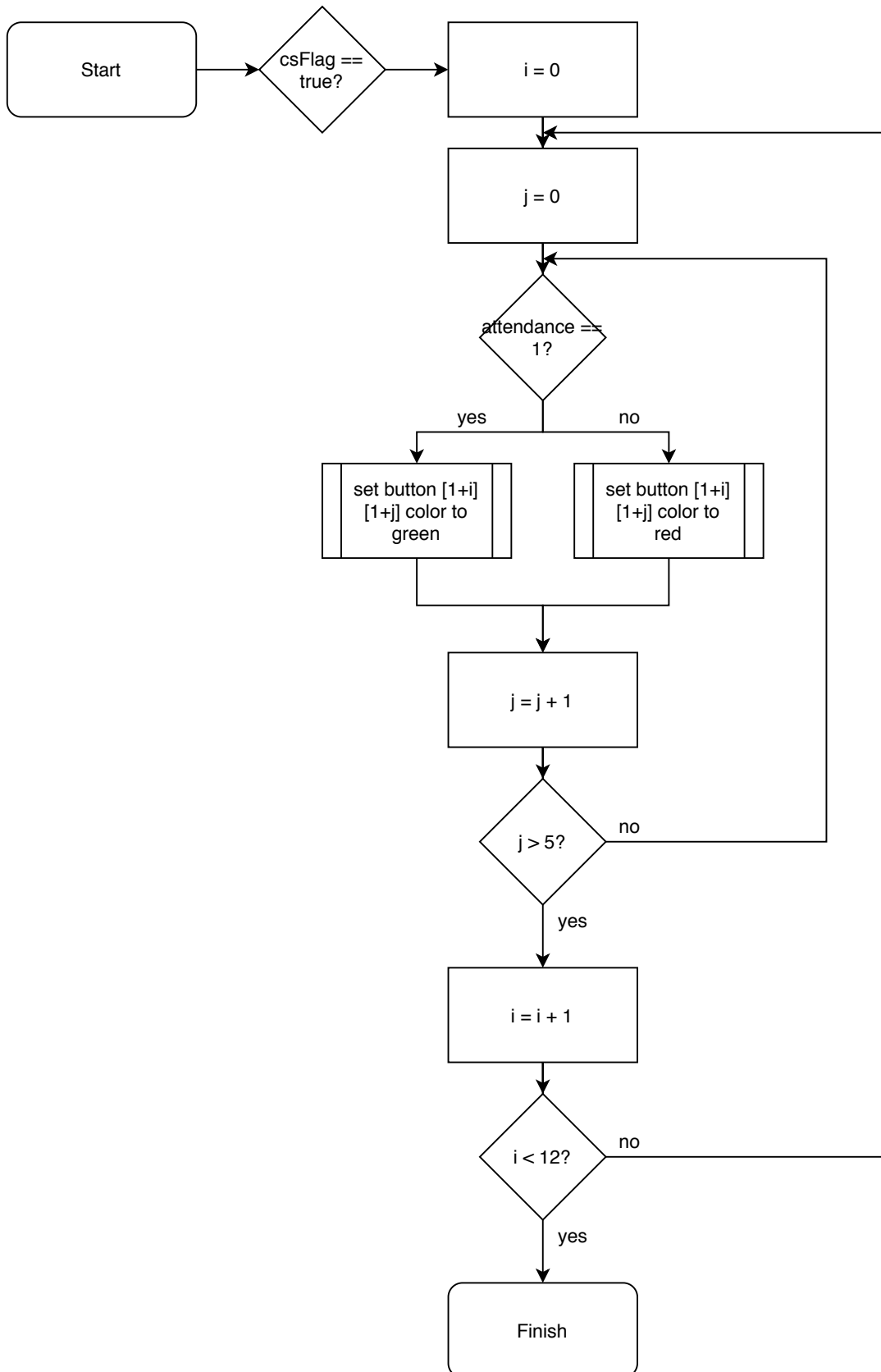
6	Check that it's possible to add a test to the calendar	After clicking a button in the calendar section it should be possible to set the name, subject and group of the test	Add the tests: "Networks" in the subject "Computer Science" for the previously created group "group 1", and "Revision test" in "Mathematics" for the group "1A".
7	Check that it's possible to edit and remove a test in the calendar	The subject, group and name of the test should be editable. The test should be removable.	Delete the test "Revision test". Edit the "Networks" test - set name to "Painting", subject to "Art" and group to "1A".
8	Marks for tests can be added, and viewed	In the "mark test" panel, marks for a test could be added, then viewed in the "all marks" panel.	Add marks for group "1A" for "Painting" test with the following marks: Anne - 1, Lucy - 2, Veronica - 3, Robert - 4, James - 5, John - 6. View the marks in the "all marks" panel.
9	After opening the program, changes made previously are saved. Check if file i/o works.	Check if elements added in one session are still visible in the next one	See if Anne's absence and the marks added are still there
10	Check if login details are encrypted	The hash tables stored in the files should not make it possible to guess the login and password	Open the files "login.txt" and "password.txt" and see what the text there looks like

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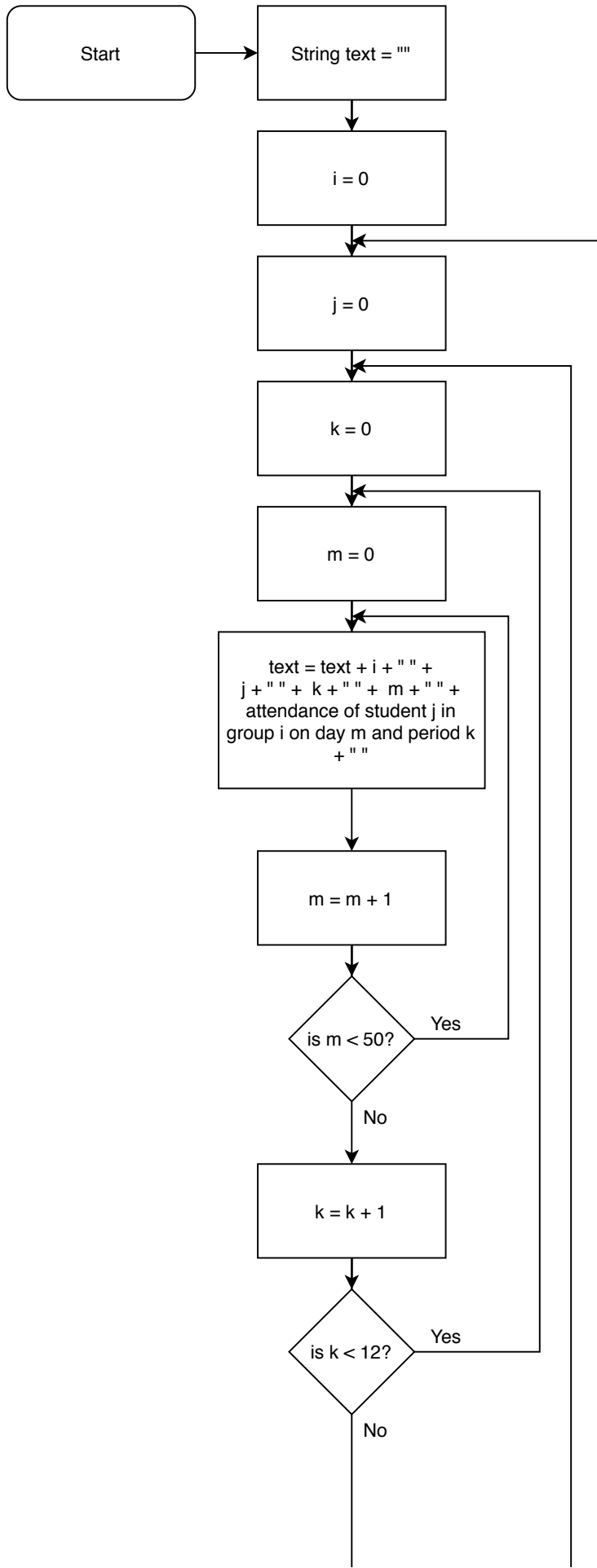
UML diagram (layout of the program).

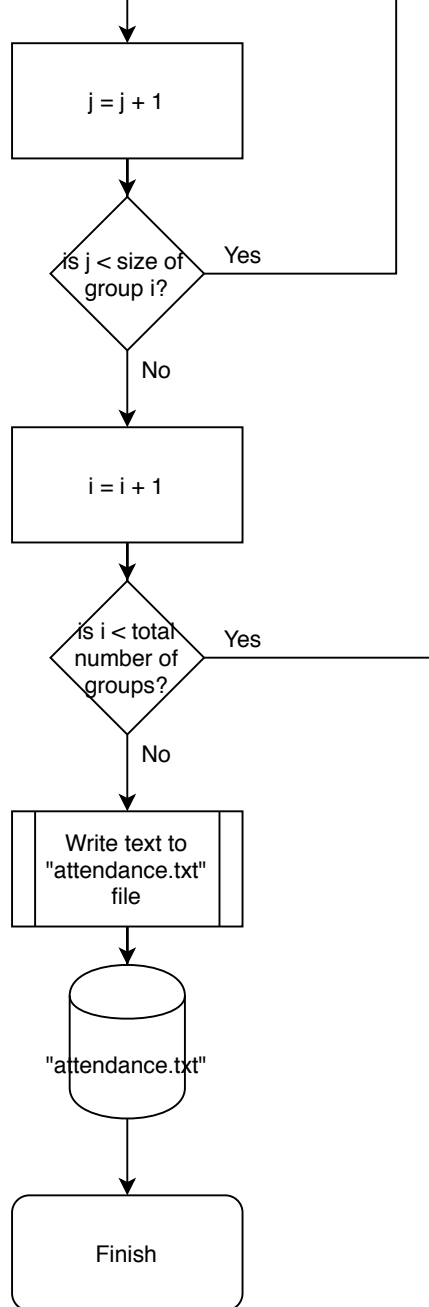


Algorithm design: algorithm changing color of the buttons in the attendance panel

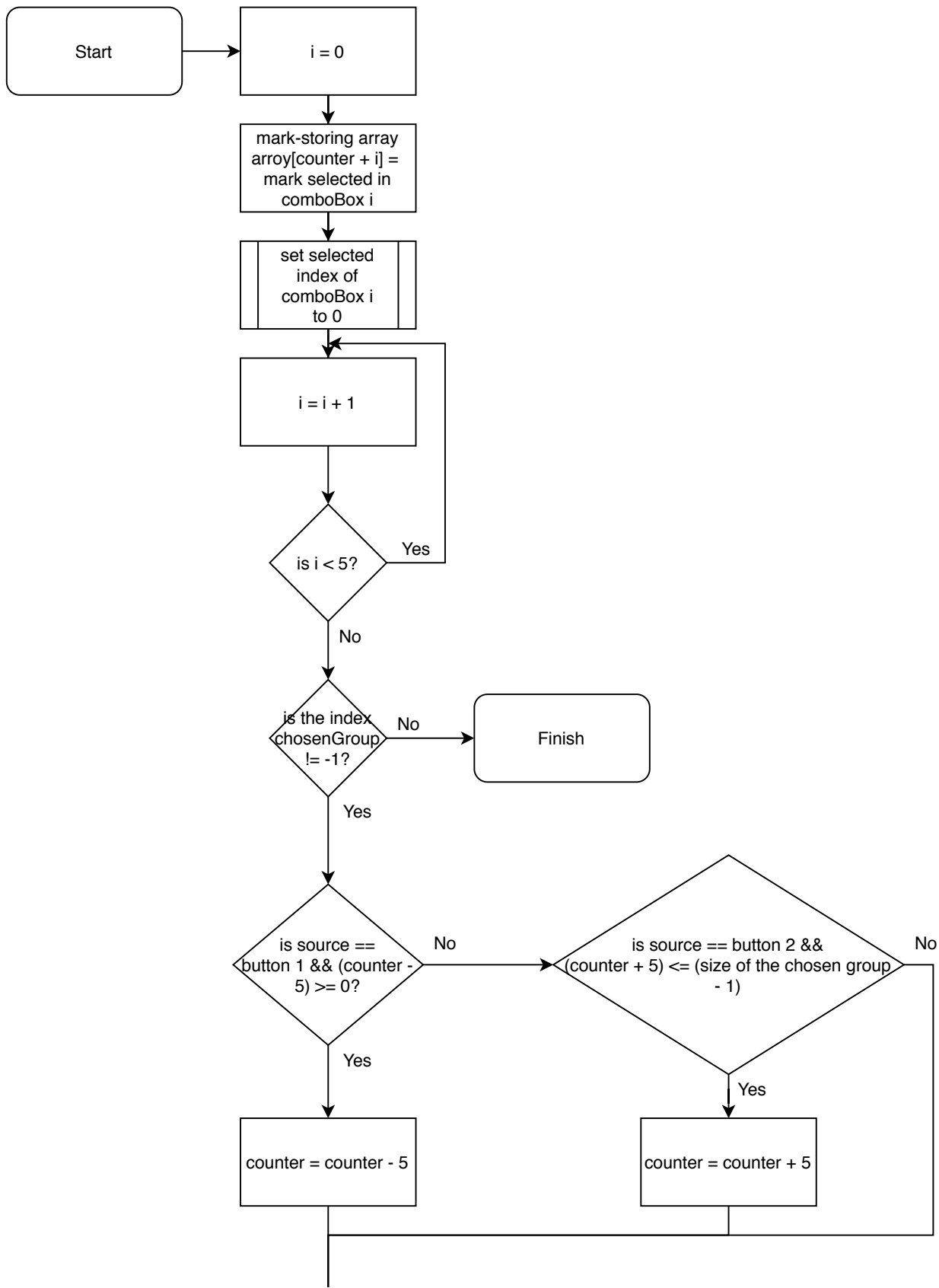


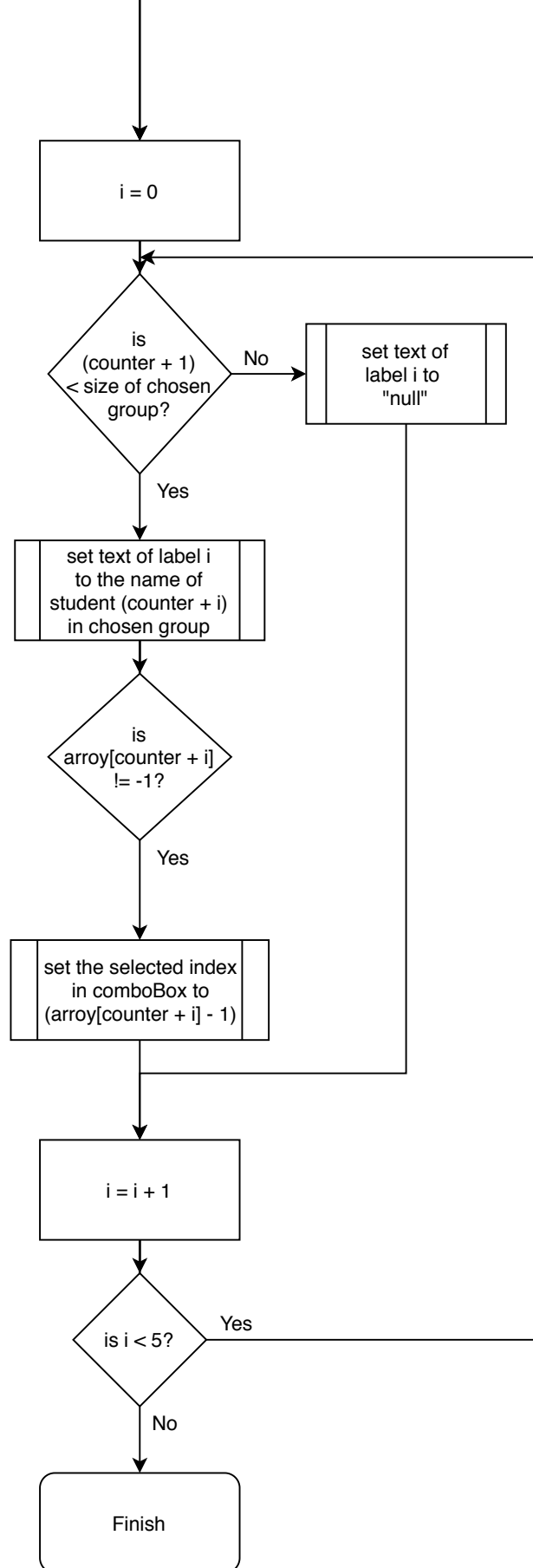
Algorithm design: algorithm saving attendance to file "attendance.txt"





Algorithm design: algorithm changing the labels (names) and comboBoxes selections (marks) in the "mark test" panel





Flowchart

